Information Spillovers in Export Destinations

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Abstract

We study how information spillovers —learning about new destinations through export activity —shape firms' entry and exit decisions in foreign markets. Using Mexican customs data, we document the existence of these spillovers and show that firms leverage their export experience to refine subsequent market choices. To rationalize these patterns, we develop a dynamic model of export supply with learning, where firms select destinations based on beliefs about profitability. We quantify the model by estimating fixed costs and foreign markets' export profitability, and use counterfactual simulations to assess the role of spillovers in shaping destination diversification. The results indicate that, absent spillovers, the share of Mexican firms exporting beyond the USA would fall by 21.6%. Moreover, reducing fixed costs in a given country not only raises entry into that market but also stimulates exporting to other destinations.

JEL Codes: D83, F12, F13

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1 Introduction

A firm's export supply depends on several factors, including productivity and trade costs. In environments with imperfect information, perceived profitability in potential destinations is also crucial. Demand uncertainty may prevent profitable entry or, conversely, induce entry followed by exit once unprofitability is revealed —an exporting failure. If such uncertainty exists, firms necessarily learn through exporting. The key question, however, is whether this learning occurs only from direct experience in a given destination or also from activity in related markets. This paper addresses precisely that question: how do information spillovers shape firms' entry and exit decisions in export markets.

The literature on firms' geographic export expansion highlights two main mechanisms. First, exporting tenure reduces entry costs, thereby raising the profitability of entering additional destinations. Second, information spillovers across destinations allow firms to leverage prior experience when selecting new markets. This paper focuses on the latter. We first provide empirical evidence that such spillovers shape firms' export dynamics. We then develop a structural model of export supply with learning, motivated by these findings, to examine the mechanisms driving the geographic spread of exports. Further research is needed to better understand how destination-specific characteristics influence entry into other markets and to design empirical models that capture firms' export choices under uncertainty.

Consider a Mexican firm evaluating entry into Germany and Switzerland but uncertain about its profitability in either market. Based on its prior beliefs, the firm enters Germany and, through exporting there, learns about its true profitability in that country. Because Germany and Switzerland share similar characteristics, the firm also updates its beliefs about Switzerland. This cross-market learning —where experience in one destination informs profitability assessments in related destinations —is what we term information spillovers. If this mechanism holds, such spillovers can help explain the geographic expansion of a firm's exports.

This paper relates to three strands of the literature. First, it contributes to work on uncertainty and entry choices, such as Das et al. (2007), Impullitti et al. (2013), and Dickstein and Morales (2018). Second, it builds on the extended gravity literature, including Morales et al. (2019), which emphasizes the role of export experience in shaping exporter dynamics. Third, it connects to research on firms' destination choices, such as Nguyen (2012), where demand uncertainty across destinations is assumed but resolved immediately once a firm enters a market. That framework, however, is static and limited to two countries. By contrast, we propose a dynamic framework in which demand shocks are imperfectly correlated across destinations, and profitability is revealed gradually through a learning process —an approach more consistent with the stylized facts for new exporters documented in Ruhl and Willis (2017). Another key distinction is methodological: whereas prior work relies on numerical simulations to assess the average number of destinations entered, we structurally estimate a model and conduct counterfactual simulations to quantify the impact of information spillovers.

Albornoz et al. (2012) study a two-period, two-country model with perfectly correlated demand shocks and immediate learning. In contrast, this paper considers an environment with multiple potential destinations where demand is only imperfectly correlated. Defever et al. (2015) provide reduced-form evidence that firms are more likely to enter a destination geographically close to one they have already served. Similarly, Evenett and Venables (2002) examine the role of past export experience, but focus exclusively on its effect on firms' cost structures. In their framework, firms are forward-looking in the sense that entry decisions account for future profitability, but not for the informational value of exporting —that is, the fact that entering a new market may generate learning about other destinations.

Schmeiser (2012) also examine the geographic expansion of firms' exports, but through a different mechanism. In their framework, expansion is driven not by demand uncertainty but by learning to export, whereby the fixed costs of entering a destination decline with the number of markets previously served. As a result, destinations that were initially unprofitable may become profitable once sufficient experience is accumulated, leading to geographic expansion. This mechanism contrasts with the one we study: in our framework, firms expand not because fixed costs fall with experience, but because exporting to one destination generates information about profitability in other markets.

The remainder of the paper is structured as follows. Section 2 describes the data, introduces our measure of destination similarity, and presents reduced-form evidence on information spillovers and their impact on firms' entry and exit decisions. Section 3 develops the model of export supply with learning, outlining how firms choose destinations and update beliefs based on experience. Section 4 details the estimation strategy, and Section 5 reports counterfactual simulations assessing the role of spillovers. Section 6 concludes with a discussion of the main findings and policy implications.

2 Empirical Evidence

This paper draws on Mexican customs data covering the universe of exporting firms. Using these data, we construct a similarity index between destinations and present reduced-form evidence on the existence of information spillovers and their effects on firms' export behavior. The most prominent empirical contribution in this area is Defever et al. (2015), who test whether exporting to one country increases the likelihood of entering a similar destination —for instance, whether prior entry into Germany raises the probability of entry into Switzerland. While this approach is intuitive and has become standard in the literature, our paper adopts a different strategy.

Unlike the studies cited above, we do not assume that export experience always raises the probability of entering similar destinations. Prior entry may simply lead to better-informed decisions. For example, a firm exporting to Germany may learn that exporting to Switzerland would not be profitable, in which case experience in Germany reduces the likelihood of entry into Switzerland. In this view, the role of export experience is informational: it reveals whether another destination is profitable. Furthermore, if exporting to one market provides information about others, that informational link may itself encourage firms to enter the destination that generates the information.

2.1 Data

The data come from the World Bank's Exporter Dynamics Database, compiled by Fernandes et al. (2015). For Mexico, the database provides annual information on export destination, sales value, year, product (at the HS 6-digit level), and, in some cases, quantity. As shown in Table 1, the sample spans 2000-2012 and includes 203,869 distinct firms exporting to 226 destinations and 2,879 unique markets. A market is defined as a destination-product pair, where products are aggregated to the HS 2-digit level.

Table 1 summarizes key exporter characteristics. On average, firms serve 2.08 destinations per year (standard deviation 3.63), and 2.54 destinations over the full sample period (standard deviation 4.57). These low averages highlight the limited diversification of Mexican exports. Indeed, exports are highly concentrated, with the United States alone accounting for 82.63% of total export value during this period. The medians, reported in parentheses, reinforce this pattern: half of all Mexican exporters serve only a single destination over their entire tenure.

Regarding tenure in a specific export market, firms export on average for only 1.93 years (s.d. 1.60). This short duration indicates that most firms experience exporting failures, entering a market briefly before exiting. Average overall tenure as an exporter is similarly low at 2.36 years (s.d. 1.99), consistent with evidence that most firms engage in international trade only fleetingly. While often attributed to adverse demand shocks, we interpret this pattern instead as a manifestation of uncertainty in export markets —an interpretation that motivates the analysis in this paper.

Statistic	Value
Years	2000 - 2012
# of firms	201,739
# of destinations	226
# of markets	2,879
Average # of destinations served per period	2.04
	(1.00)
Average # of markets served per period	3.38
	(1.00)
Average # of destinations served in total	2.47
	(1.00)
Average # of markets served in total	4.57
	(2.00)
Average tenure of a firm in an export destination	1.87
	(1.00)
Average tenure of a firm in an export market	1.67
	(1.00)
Average tenure of a firm as an exporter	2.25
	(1.00)

Table 1: Descriptive Statistics.

2.2 Similarity of Export Destinations

To construct a similarity index between export destinations, we follow Finger and Kreinin (1979), who developed an export similarity index. This index captures how similar two countries are in terms of their export patterns to a common third market and is defined as:

$$S_X(i,j;k) = \sum_l \min\left(\frac{X_{ik}^l}{X_{ik}}, \frac{X_{jk}^l}{X_{jk}}\right) \tag{1}$$

In this notation, i and j denote the two origin countries, k the third country, l a trade classification category, and X the value of exports. For our purposes, however, we adapt this framework to construct an import-based similarity index. We use imports rather than exports because Mexican firms care about how similar potential destinations are in terms of their demand for Mexican goods —that is, how similar they appear from the exporter's perspective. Formally, the import-based similarity index is defined as:

$$S(i,j;k) = \sum_l \min\left(\frac{M_{ik}^l}{M_{ik}}, \frac{M_{jk}^l}{M_{jk}}\right) \tag{2}$$

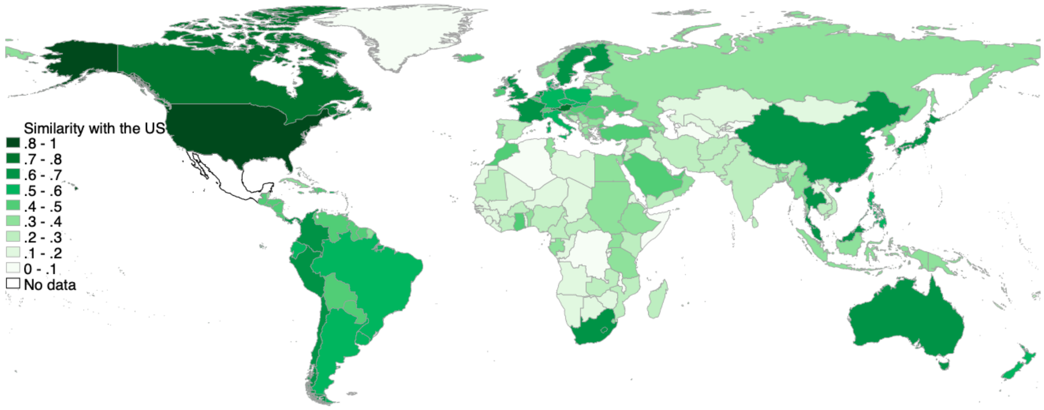


Figure 1: The US compared to the ROW in terms of their imports from Mexico.

The index takes values in the interval $S \in [0, 1]$, where higher values indicate greater similarity between countries i and j in terms of their imports from k. An index of 0 implies entirely dissimilar import patterns, while a value of 1 implies identical commodity distributions. To compute the import-based similarity index, we use Mexican customs data, which allow firm exports to be aggregated at the product-destination level. This enables us to measure, for each product, the value of imports from Mexico by every destination country. Products are classified according to the HS system at the 2-digit level.

Figure 1 illustrates the similarity between the United States and other countries in terms of their imports from Mexico. The index produces intuitive patterns, such as high similarity with Canada and with Commonwealth countries including the United Kingdom, Australia, and South Africa. The final outcome of this exercise is a symmetric matrix capturing similarity across all country pairs for destinations served by Mexican exporters between 2000 and 2012. Although the index is computed annually, we average it across years for use in the reduced-form estimations.

2.3 Identification Strategy

This section presents empirical evidence on the existence of information spillovers and their impact on firms' export market choices. If destinations provide information about other potential markets, two testable implications follow:

Implication 1: *If destinations convey information about other similar destinations, then firms with prior experience in countries similar to i should be less likely to make incorrect entry decisions in i.*

Assuming uncertainty in destination-specific profitability and the presence of information spillovers, firms with greater export experience in similar markets are less likely to make incorrect entry decisions into a given destination.

Implication 2: *If destinations provide information about other similar destinations, their value lies in the information they reveal about additional markets. Destinations that are more informative should therefore be more likely to be entered.*

With information spillovers, and assuming firms are aware of them and behave forward-looking, destinations are valuable not only for their expected profitability but also for the information they provide about untapped markets. A firm may choose to enter a destination even when expected profits are non-positive if doing so reduces uncertainty about other markets, thereby improving future profitability or at least enabling better entry decisions.

2.3.1 Relevant Export Experience and Incorrect Entry Decisions

Ideally, identification would rely on regressing the probability of an "incorrect" entry decision into a destination on a measure of the information a firm holds about that destination, using export experience as a proxy for information. In practice, however, defining or estimating what constitutes an incorrect decision is difficult —particularly given the limitations of customs data, which lack firm-level characteristics.

Testing this implication requires a clear definition of an incorrect decision. We define it as the case in which a firm enters a market, later finds exporting unprofitable, and subsequently exits —that is, entry when entry was not profitable. This, in turn, requires defining what constitutes exit from a market and how to proxy for profitability. For exit, we distinguish between two cases:

- An exit occurs when a firm enters a market and exits in the following period.
- An exit occurs when a firm enters and subsequently exits at any point prior to the final year of the sample.

Measuring profitability in an export market is more challenging. Since our data lack firm-level characteristics, we proxy profitability by comparing firms' exit or continuation decisions with demand conditions: firms that either exit when demand is rising or continue when demand is falling. Formally:

- A firm is classified as profitable if it remains in a market despite a negative demand shock.

- A firm is classified as unprofitable if it exits a market despite a positive demand shock.

The rationale is as follows: if a firm exits a market despite a positive demand shock, this suggests it was not profitable and therefore chose to leave. Conversely, if a firm remains in a market despite a negative demand shock, this indicates that it was likely profitable, since it was able to withstand adverse demand conditions.

Finally, we define positive and negative demand shocks based on changes in demand. Although our analysis is conducted at the market level (a product-destination pair), we measure demand shocks at the destination level. This avoids reverse causality: if only a few firms serve a market, one firm's exit could itself generate a negative market shock. We consider two alternative measures of a positive demand shock:

- The growth rate of demand is positive.

- The growth rate of demand is above its average growth rate.

An equivalent definition applies for negative demand shocks. Using these definitions, Table 2 illustrates how we construct the dependent variable. We exclude cases in which a firm exits under a negative demand shock or continues under a positive demand shock, since profitability is ambiguous in these situations. For example, if a firm exits during a negative demand shock, this decision may simply reflect deteriorating demand rather than low firm-specific profitability. Likewise, if a firm continues during a positive demand shock, persistence cannot be taken as evidence of profitability.

	Positive demand shock	**Negative demand shock**
Exit	Incorrect decision	-
Continue	-	Correct decision

Table 2: Definition of a incorrect decision.

We now turn to the independent variable of interest: the measure of relevant export experience. We use the term relevant to highlight that not all previously served destinations provide the same amount of information about a given market; this depends on their similarity to the destination under consideration. The greater the similarity between a firm's prior destinations and the market in question, the more information the firm possesses, and the more likely it is to make an informed entry decision.

We define relevant export experience as a function of two components: the set of destinations served by firm i prior to period t, and the degree of similarity between each of those destinations and market m. Let z_{imt} denote the export experience of firm i with respect to market m at time t. Formally:

$$z_{imt} = \sum_{j \in V_{i,t-1}} S(m, j) \tag{3}$$

where the summation is taken over all destinations $j \in V_{i,t-1}$, the set of markets served by firm i up to period $t-1$. By construction, z_{imt} increases with the size of $V_{i,t-1}$ whenever $S(m, j) > 0$ —that is, when previously served destinations are at least somewhat similar to m. The corresponding specification to test whether greater information improves entry decisions is:

$$\mathbb{P}_{imt} = \alpha_0 + \alpha_1 z_{imt} + \epsilon_{imt} \tag{4}$$

where $\mathbb{P}_{imt}$ denotes the probability that firm i makes an incorrect entry decision in market m at time t, as defined by the criteria above. We estimate a linear probability model to examine how relevant export experience influences the likelihood of making such a decision.

The specification in Equation (4) faces potential identification challenges, as other factors may also influence entry into export markets. Two concerns are particularly relevant. First, positive demand shocks could simultaneously drive entry into multiple destinations, confounding the estimated effect of export experience. For instance, a shock affecting both Germany and Switzerland —possibly with different timing —might induce entry into Germany and later into Switzerland. In this case, the coefficient on export experience would be upward biased, overstating its true effect on entry decisions.

The second concern is firm-level learning about how to export. Prior research shows that firms become more effective exporters as they accumulate experience. If so, destination profitability would rise with overall export experience, again confounding the identification of information spillovers. For example, a firm may enter Switzerland after exporting to Germany not because Germany revealed information about Switzerland's profitability, but because exporting to Germany improved the firm's efficiency in exporting, which in turn facilitated entry into Switzerland.

To control for both of these concerns, we propose the following specification:

$$\mathbb{P}_{imt} = \alpha_0 + \alpha_1 z_{imt} + \eta_1 \%\Delta x_{dt} + \eta_2 \%\Delta x_{it} + \phi_{f,m,t} + \epsilon_{imt} \tag{5}$$

where $\%\Delta x_{dt}$ denotes the percentage change in demand in destination d at time t, and $\%\Delta x_{it}$ the percentage change in firm i's aggregate exports at time t. We also include product, market, and time fixed effects to account for unobserved heterogeneity. The variable $\%\Delta x_{dt}$ controls for destination-specific demand shocks: if a shock occurs in d at time t, it will affect all firms exporting there and thus be captured by this term. Meanwhile, $\%\Delta x_{it}$ controls for firm-level learning-to-export effects and quality upgrading: if accumulated experience increases efficiency, it should raise exports across all destinations served by the firm, which is reflected in changes in its aggregate exports.

In Table 3 each column corresponds to a different definition of an incorrect decision. The dependent variable $\mathbb{P}_{x,y}$ denotes the probability of making an incorrect decision under criteria (x, y), where x specifies the timing of exit and y the definition of a positive demand shock. Specifically, $x = inm$ indicates immediate exit from a market, $y = pdg$ defines a positive demand shock as positive demand growth, and $y = dgaa$ defines it as above-average demand growth. Alternative combinations of these definitions can also be examined as robustness checks.

	(1) $\mathbb{P}_{inm,pdg}$	(2) $\mathbb{P}_{inm,pdg}$	(3) $\mathbb{P}_{inm,dgaa}$
Export Experience	-0.00527*** (-18.41)	-0.0118*** (-5.78)	-0.0120*** (-5.34)
$\%\Delta x_{dt}$		-1.85e-08 (-0.79)	-7.93e-08 (-1.46)
$\%\Delta x_{it}$		1.22e-09* (2.84)	1.87e-09** (3.23)
Constant	0.187*** (81.78)	0.216*** (21.79)	0.435*** (51.05)
Fixed Effects		✓	✓
Observations	40,984	25,561	17,922

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3: Relevant Export Experience.

Table 3 reports results from both the baseline specification in Equation (4), which excludes controls and fixed effects, and the full specification in Equation (5). The estimated coefficient on relevant export experience appears biased when other determinants of entry are not accounted for. Columns (2) and (3) present the full specification under alternative definitions of an incorrect decision. The results indicate that the coefficient on export experience is robust to these alternative definitions.

The coefficient on relevant export experience is significant and has the expected sign: greater relevant experience reduces the likelihood of making an incorrect entry decision. Quantitatively, a one-standard deviation increase in relevant experience (5.4 points) lowers the probability of an incorrect decision by 6.4%. In the last two specifications, standard errors are clustered at the firm, market, and time levels to account for correlation in unobservables across these dimensions. The reduction in observations when including controls reflects the fact that some destinations are not served in contiguous years, coupled with many firms not exporting in consecutive periods. Finally, the relatively large coefficients on the control variables arise from substantial demand growth rates, driven by the high turnover observed in some export markets.

In sum, the evidence indicates that firms acquire information about other potential destinations through exporting and use it to improve their entry decisions. This finding supports the hypothesis stated in **Implication 1**.

2.3.2 Informativeness of Destinations and Entry

The objective of this section is to test whether the informativeness of a destination influences the probability of entry —that is, whether forward-looking firms enter certain markets partly to acquire information about others. To this end, Equation 6 defines our measure of destination informativeness. For firm i at period t, the index captures how informative destination d is by considering the set of destinations not yet served by the firm and their similarity to d. Formally:

$$I_{idt} = \sum_{j \notin V_{i,t-1} \cup \{d\}} S(d, j) \tag{6}$$

Using the same notation as in Section 2.3.1, except that the summation now applies to destinations not previously served by firm i, we test the following relationship:

$$\mathbb{P}[\text{entry}_{idt}] = \beta_0 + \beta_1 I_{idt} + \epsilon_{idt} \tag{7}$$

We expect the sign of the coefficient for informativeness to be positive, representing that the more informative a destination is for a firm, the higher the probability the firm will enter that destination, given that it would allow it to learn about other potential markets. As with the analysis for the effect of relevant export experience, the specification in 7 could be misleading as other factors can potentially affect the probability of entry into a destination. One of these factors, as discussed in Section 2.3.1, is demand shocks driving entry into a destination, i.e., if a firm entered a given destination, it might be just because of a positive destination demand shock, which now made the destination profitable.

To address this concern, we estimate the specification in Equation (8). In this framework, changes in profitability —proxied by positive demand shocks —are captured by the variable $\%\Delta x_{dt}$. In addition, we include destination and time fixed effects to absorb other unobserved sources of variation in entry decisions.

$$\mathbb{P}[\text{entry}_{idt}] = \beta_0 + \beta_1 I_{imt} + \eta_1 \%\Delta x_{dt} + \phi_{d,t} + \epsilon_{idt} \tag{8}$$

Table 4 reports results for a random sample of firms. Column (1) presents the baseline specification in Equation (7), Column (2) adds the control variable but excludes fixed effects, and Column (3) reports the full specification from Equation (8).

	(1) $\mathbb{P}(entry)$	(2) $\mathbb{P}(entry)$	(3) $\mathbb{P}(entry)$
Informativeness	0.0000577***	0.0000782***	0.00338***
	(166.70)	(160.07)	(10.93)
$\%\Delta x_{dt}$		7.67e-10***	2.52e-09*
		(124.68)	(2.37)
Constant	-0.00152***	-0.00265***	-0.163***
	(-151.87)	(-145.95)	(-10.85)
Fixed Effects			✓
Observations	33,132,504	27,671,505	27,671,505

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4: Informativeness Index.

Focusing on the results from the full specification in Column (3), the coefficient on our informativeness measure is positive, consistent with the hypothesis that more informative destinations are more likely to be entered. The magnitude of the effect is also reasonable: a one-standard deviation increase in informativeness (20.06 points) raises the probability of entry by 6.8%. The standard errors in Columns (1) and (2) highlight the importance of clustering, as they reveal correlation across unobservables. Accordingly, in Column (3) we cluster standard errors at both the firm and destination levels. As in Section 2.3.1, the number of observations falls once controls are included, reflecting the lack of contiguous years for some destinations in our random sample and the high turnover of exporters.

In summary, the evidence supports **Implication 2**: if destinations are informative about other potential markets, and if firms are forward-looking, then more informative destinations are more likely to be entered. Firms value this information because it enables better future entry decisions.

3 A Model of Export Supply and Information Spillovers

The most prominent models of information spillovers and firm entry into export markets are developed by Nguyen (2012), Albornoz et al. (2012), and Morales et al. (2019). Our framework follows Cebreros (2016) but extends it to multiple destinations and allows for information spillovers across them. The model builds on the standard heterogeneous-firm framework of Melitz (2003), incorporating uncertainty about destination-specific demand shocks. Exporters choose destinations discretely based on prior beliefs, which they update using Bayes' Rule.

3.1 Consumer Demand at Export Destinations

On the demand side, we assume that utility in export destination j is described by CES preferences:

$$U_j = \left(\int_{\Omega} \epsilon_j(\omega) q_j(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}} \tag{9}$$

where $\sigma > 1$ denotes the elasticity of substitution across varieties, $q_j(\omega)$ is consumption of variety ω, and $\epsilon_j(\omega)$ is a demand shifter. Given aggregate expenditure X_j, the demand faced by firm ω in country j is:

$$q_j(\omega) = \epsilon_j(\omega)^{\sigma} p_j(\omega)^{-\sigma} X_j P_j^{\sigma-1} \tag{10}$$

where P_j is the Dixit-Stiglitz price index:

$$P_j \equiv \left(\int_{\Omega} \epsilon_j(\omega)^{\sigma} p_j(\omega)^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \tag{11}$$

Using $r(\omega) \equiv p_j(\omega) q_j(\omega)$ to denote revenue, we can substitute the demand expression to rewrite exporter ω's revenue in market j as:

$$r(q_j) = \epsilon_j(\omega) X_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}} q_j^{\frac{\sigma-1}{\sigma}} \tag{12}$$

We assume that all uncertainty in export market j is captured by underlying profitability θ_{jt}, which is unknown to the firm. Accordingly, we rewrite the demand shifter as $h(\theta_{jt})$, encompassing both aggregate determinants of demand, $X_j^{\frac{1}{\sigma}} P_j^{\frac{\sigma-1}{\sigma}}$, and the idiosyncratic component $\epsilon_j(\omega)$. For simplicity, we assume no uncertainty in the domestic market and normalize its demand shifter to unity. Under these assumptions, firm revenue is given by:

$$\text{Domestic Revenue: } r(q_j) = q_j^{\frac{\sigma-1}{\sigma}} \tag{13}$$

$$\text{Export Revenue: } r(q_j) = h(\theta_{jt}) q_j^{\frac{\sigma-1}{\sigma}} \tag{14}$$

3.2 Static Exporter Behavior

For now, we describe the firm's profit maximization problem as static. Conditional on the vector θ_t, total revenue is the sum of revenues across all markets:

$$r_t = q_{ht}^{\frac{\sigma-1}{\sigma}} + \sum_{j \in D} d_{jt} h(\theta_{jt}) q_{jt}^{\frac{\sigma-1}{\sigma}} \tag{15}$$

where:

$$d_{jt} = \begin{cases} 1 & \text{if the firm exports to destination } j \text{ in period } t, \\ 0 & \text{otherwise.} \end{cases}$$

where q_{ht} and q_{jt} denote quantities sold in the domestic market and in export market j, respectively. Conditional on export status $d_t \in \mathbb{R}^D$, profit-maximizing firms equate marginal revenue across the domestic and all export markets:

$$q_{jt} = d_{jt} h(\theta_{jt})^{\sigma} q_{ht} \quad \forall j \in D \tag{16}$$

we can therefore define the firm's total output as:

$$q_t = \Big(1 + \sum_{j \in D} d_{jt} h(\theta_{jt})^{\sigma}\Big) q_{ht} \tag{17}$$

Equations (16) and (17) allow us to rewrite a firm's total revenue as:

$$
\begin{aligned}
r_t &= q_{ht}^{\frac{\sigma-1}{\sigma}} + \sum_{j\in D} d_{jt}h(\theta_{jt})[d_{jt}h(\theta_{jt})^{\sigma} q_{ht}]^{\frac{\sigma-1}{\sigma}} = q_{ht}^{\frac{\sigma-1}{\sigma}} + \sum_{j\in D} d_{jt}h(\theta_{jt})^{\sigma} q_{ht}^{\frac{\sigma-1}{\sigma}} \\
&= \Big(1+\sum_{j\in D} d_{jt}h(\theta_{jt})^{\sigma}\Big) q_{ht}^{\frac{\sigma-1}{\sigma}} = \Big(1+\sum_{j\in D} d_{jt}h(\theta_{jt})^{\sigma}\Big)\Big[\Big(1+\sum_{j\in D} d_{jt}h(\theta_{jt})^{\sigma}\Big)^{-1} q_t\Big]^{\frac{\sigma-1}{\sigma}} \\
&= \Big(1+\sum_{j\in D} d_{jt}h(\theta_{jt})^{\sigma}\Big)^{\frac{1}{\sigma}} q_t^{\frac{\sigma-1}{\sigma}} \qquad (18)
\end{aligned}
$$

Let f_j denote the fixed cost of exporting to destination j, paid in each period the firm serves that market. Conditional on θ_t and export status d_t, firms choose total output to maximize profits:

$$
\max_{q_t} \left\{ \Big(1+\sum_{j\in D} d_{jt}h(\theta_{jt})^{\sigma}\Big)^{\frac{1}{\sigma}} q_t^{\frac{\sigma-1}{\sigma}} - \Big(f+\sum_{j\in D} d_{jt}f_j\Big) - \Big(1+\sum_{j\in D} \tau_{jt}d_{jt}h(\theta_{jt})^{\sigma}\Big)\Big(1+\sum_{j\in D} d_{jt}h(\theta_{jt})^{\sigma}\Big)^{-1} q_t \right\} \qquad (19)
$$

where the first term in parentheses represents total revenue, the second the fixed costs of serving domestic and export markets, and the third the firm's variable costs. The only source of heterogeneity across firms is export profitability. We normalize the marginal cost of serving the domestic market to unity, so the marginal cost of serving export market j is given by τ_{jt}.

Solving the firm's maximization problem yields the following expression for its optimal scale of operation:

$$
q_t = \Big(\frac{\sigma-1}{\sigma}\Big)^{\sigma}\Big(1+\sum_{j\in D} \tau_{jt}d_{jt}h(\theta_{jt})^{\sigma}\Big)^{-\sigma}\Big(1+\sum_{j\in D} d_{jt}h(\theta_{jt})^{\sigma}\Big)^{1+\sigma} \qquad (20)
$$

From this, we derive an expression for profits as a function of export status d_t, conditional on export profitability:

$$
\Pi(d_t|\theta_t) = \frac{1}{\sigma}\Big(\frac{\sigma-1}{\sigma}\Big)^{\sigma-1}\Big(1+\sum_{j\in D} \tau_j d_{jt}h(\theta_{jt})^{\sigma}\Big)^{1-\sigma}\Big(1+\sum_{j\in D} d_{jt}h(\theta_{jt})^{\sigma}\Big)^{\sigma} - \Big(f+\sum_{j\in D} d_{jt}f_j\Big) \qquad (21)
$$

Given that the firm has uncertainty concerning its export profitability θ_t, the firm's expected profits are given by:

$$\tilde{\Pi}(d_t|\Gamma_t) = \frac{1}{\sigma}\Big(\frac{\sigma-1}{\sigma}\Big)^{\sigma-1}\mathbb{E}_{\theta_t}\Big[\Big(1+\sum_{j\in D}\tau_j d_{jt}h(\theta_{jt})^{\sigma}\Big)^{1-\sigma}\Big(1+\sum_{j\in D}d_{jt}h(\theta_{jt})^{\sigma}\Big)^{\sigma}|\Gamma_t\Big] - \Big(f+\sum_{j\in D}d_{jt}f_j\Big) \quad (22)$$

Let Γ_t denote the firm's information set in period t, determined by its prior export experience. Equation (22) is central, as it expresses expected profits as a function of export status d_t given Γ_t. In each period, the firm chooses its export status $d_t = [d_{1t}, \ldots, d_{Dt}]$ to maximize expected profits conditional on Γ_t. Defining:

$$A_t(d_t|\Gamma_t) \equiv \mathbb{E}_{\theta_t}\Big[\Big(1+\sum_{j\in D}\tau_j d_{jt}h(\theta_{jt})^{\sigma}\Big)^{1-\sigma}\Big(1+\sum_{j\in D}d_{jt}h(\theta_{jt})^{\sigma}\Big)^{\sigma}|\Gamma_t\Big]^{\frac{1}{\sigma}} \quad (23)$$

Results in the final expression for a firm's expected profits given d_t:

$$\tilde{\Pi}(d_t|\Gamma_t) = \frac{1}{\sigma}\Big(\frac{\sigma-1}{\sigma}\Big)^{\sigma-1}A_t(d_t)^{\sigma} - \Big(f+\sum_{j\in D}d_{jt}f_j\Big) \quad (24)$$

where $A_t(d_t)$ captures all uncertainty about a firm's export profitability across markets. This expression reflects a static optimization problem, as it ignores the dynamic implications of entry choices. Yet dynamics are essential: if information spillovers exist and firms are forward-looking, a firm may enter a destination not only for its expected profits but also for the information it provides about other markets.

3.3 Learning and Information Spillovers

This section describes how firms acquire information and update beliefs using Bayes' Rule. Assuming firms know the functional form of $h(\cdot)$, when a firm exports to destination j and observes realized demand in period t, it receives a noisy signal of its export profitability in that market:

$$s_{jt} = \theta_j + \epsilon_{jt} \quad \text{where} \quad \epsilon_{jt} \overset{\text{i.i.d.}}{\sim} \mathbb{N}(0, \sigma_{\epsilon}^2) \quad (25)$$

We assume the vector of true export profitability, $\theta \in \mathbb{R}^D$, follows a multivariate normal distribution $\mathbb{N}(\mu_\theta, \Sigma_\theta)$. Firms are assumed to know both μ_θ and Σ_θ, as well as the distribution of ϵ_{jt}.

True export profitability is time-invariant but subject to random shocks unobservable to firms. Each period that a firm exports to destination $j \in D$, it receives a signal s_{jt} and uses it to update beliefs about the distribution of θ. Without information spillovers, only the belief regarding θ_j would be updated, since exporting to j would provide no information about other destinations. We introduce spillovers by assuming that export profitabilities $\theta_j j \in D$ are correlated, so that the covariance matrix $\Sigma\theta$ is symmetric and non-diagonal. Specifically, the covariance between θ_j and θ_k is modeled as a function of the similarity between destinations j and k.

Learning in the model unfolds as follows. At $t = 0$, each firm draws its vector θ from $\mathbb{N}(\mu_\theta, \Sigma_\theta)$. Since firms know this distribution, it represents their prior beliefs about export profitability. At $t = 1$, firms choose export status d_1 to maximize Equation (23) based on these priors. For any $t > 1$, export decisions depend on updated beliefs about profitability, which are functions of the firm's export history up to that point. The learning process for an arbitrary export history $d^t = (d_1, \ldots, d_t)$ is derived as follows.

Bayes' Rule implies that the distribution of θ conditional on observing signals s_t is proportional to the likelihood of those signals multiplied by the prior distribution of θ:

$$\pi(\theta|z_t) \propto L(s_t|\theta)\pi(\theta) \tag{26}$$

where s_t is the vector of signals observed in period t, and $\pi(\theta)$ is the (common knowledge) distribution of θ. Let $\Phi_t = j \mid d_{jt} = 1$ denote the set of destinations served by the firm in period t. Then:

$$s_{jt} = \theta_j + \epsilon_{jt} \;\; \forall j \in \Phi_t \tag{27}$$

$$\Rightarrow s_{jt}|\theta_j \sim \mathbb{N}(\theta_j, \sigma_\epsilon^2) \tag{28}$$

Conditional on θ, realizations of s_t are i.i.d. due to the distribution of error term ϵ_{jt} and therefore we can express the likelihood of s_t conditional on θ as follows:

$$\begin{aligned} L(s_t|\theta) &= \prod_{j \in \Phi_t} \mathbb{N}(\theta_j, \sigma_\epsilon^2) \\ &\propto \exp\Big(-\frac{1}{2} \sum_{j \in \Phi_t} \Big(\frac{s_{jt} - \theta_j}{\sigma_\epsilon}\Big)^2 \Big) \end{aligned} \tag{29}$$

Define $\Omega_t \equiv \text{diag}((1/\sigma_\epsilon^2)d_t)$ which is a diagonal and therefore symmetrical matrix such that $\Omega_{t_{jj}} = 0$ if $d_{jt} = 0$. We can then rewrite the likelihood above as:

$$L(s_t|\theta) \propto \exp\Big(-\frac{1}{2}(\theta_t-\theta)'\Omega_t(\theta_t-\theta)\Big) \tag{30}$$

Beliefs are updated after a firm has chosen its export status, which are the beliefs it carries into the next period. Using Equation (26), the posterior distribution of a firm's export profitability given any export history d^{t-1} can be expressed as:

$$\pi(\theta|d^{t-1}) = \pi(\theta)\prod_{i=1}^{t-1} L(s_i|\theta) \tag{31}$$

where $\pi(\theta)$ is common knowledge and thus acts as firms' prior beliefs at the beginning of the learning process. Conditional on θ, the s_i terms are i.i.d. which allows us to express the above as the product of likelihood functions and therefore:

$$\begin{aligned}
\pi(\theta|d^{t-1}) &\propto \pi(\theta)\prod_{i=1}^{t-1}\exp\Big(-\frac{1}{2}(s_i-\theta)'\Omega_i(s_i-\theta)\Big)\\
&\propto \exp\Big(-\frac{1}{2}(\theta-\mu_\theta)'\Sigma_\theta^{-1}(\theta-\mu_\theta)\Big)\times\prod_{i=1}^{t-1}\exp\Big(-\frac{1}{2}(s_i-\theta)'\Omega_i(s_i-\theta)\Big)\\
&\propto \exp\Big(-\frac{1}{2}(\theta-\mu_\theta)'\Sigma_\theta^{-1}(\theta-\mu_\theta)-\frac{1}{2}\sum_{i=1}^{t-1}(s_i-\theta)'\Omega_i(s_i-\theta)\Big)
\end{aligned} \tag{32}$$

For the sake of convenience, we take the natural logarithm of the whole expression and multiply both quadratic terms, resulting in the following:

$$
\begin{aligned}
\ln\pi(\theta|d^{t-1}) &\propto -\frac{1}{2}(\theta-\mu_\theta)^{'}\Sigma_\theta^{-1}(\theta-\mu_\theta) - \frac{1}{2}\sum_{i=1}^{t-1}(s_i-\theta)^{'}\Omega_i(s_i-\theta) \\
&= -\frac{1}{2}\Big[\theta^{'}\Sigma_\theta^{-1}\theta - \theta^{'}\Sigma_\theta^{-1}\mu_\theta - \mu_\theta^{'}\Sigma_\theta^{-1}\theta + \mu_\theta^{'}\Sigma_\theta^{-1}\mu_\theta \\
&+ \sum_{i=1}^{t-1}s_i^{'}\Omega_i s_i - \sum_{i=1}^{t-1}s_i^{'}\Omega_i\theta - \sum_{i=1}^{t-1}\theta^{'}\Omega_i s_i + \sum_{i=1}^{t-1}\theta^{'}\Omega_i\theta\Big] \\
&= -\frac{1}{2}\theta^{'}\Big(\Sigma_\theta^{-1} + \sum_{i=1}^{t-1}s_i^{'}\Omega_i\Big)\theta + \frac{1}{2}\Big(\sum_{i=1}^{t-1}s_i^{'}\Omega_i\theta + \sum_{i=1}^{t-1}\theta^{'}\Omega_i s_i\Big) \\
&+ \frac{1}{2}\Big(\theta^{'}\Sigma_\theta^{-1}\mu_\theta + \mu_\theta^{'}\Sigma_\theta^{-1}\theta\Big) - \frac{1}{2}\Big(\mu_\theta^{'}\Sigma_\theta^{-1}\mu_\theta + \sum_{i=1}^{t-1}s_i^{'}\Omega_i s_i\Big)
\end{aligned}
\tag{33}
$$

Since μ_θ, Σ_θ, Ω_i and s_i are either observed or known by the firm, we can ignore the last term in parenthesis. Moreover, because of the dimensions of these matrices, we have that $s_i^{'}\Omega_i\theta = \theta^{'}\Omega_i s_i$ and $\mu_\theta^{'}\Sigma_\theta^{-1}\theta = \theta^{'}\Sigma_\theta^{-1}\mu_\theta$ and therefore:

$$
\begin{aligned}
\ln\pi(\theta|d^{t-1}) &\propto -\frac{1}{2}\theta^{'}\Big(\Sigma_\theta^{-1} + \sum_{i=1}^{t-1}\theta_i^{'}\Omega_i\Big)\theta + \sum_{i=1}^{t-1}\theta^{'}\Omega_i s_i + \theta^{'}\Sigma_\theta^{-1}\mu_\theta \\
&= -\frac{1}{2}\theta^{'}\Big(\Sigma_\theta^{-1} + \sum_{i=1}^{t-1}s_i^{'}\Omega_i\Big)\theta + \theta^{'}\Big(\Sigma_\theta^{-1}\mu_\theta + \sum_{i=1}^{t-1}\Omega_i s_i\Big)
\end{aligned}
\tag{34}
$$

which yields the following proportional expression:

$$
\begin{aligned}
\ln\pi(\theta|d^{t-1}) &\propto -\frac{1}{2}\Big(\theta-\mu_\theta^p\Big)^{'}\Big(\Sigma_\theta^{-1} + \sum_{i=1}^{t-1}\Omega_i\Big)\Big(\theta-\mu_\theta^p\Big) \\
\text{with } \mu_\theta^p &= \Big(\Sigma_\theta^{-1} + \sum_{i=1}^{t-1}\Omega_i\Big)^{-1}\Big(\Sigma_\theta^{-1}\mu_\theta + \sum_{i=1}^{t-1}\Omega_i s_i\Big)
\end{aligned}
\tag{35}
$$

All of this results in the firm's posterior beliefs for its export profitability at the beginning of period t given export history d^{t-1} being:

$$
\theta|d^{t-1} \sim \mathbb{N}\Big(\Big(\Sigma_\theta^{-1} + \sum_{i=1}^{t-1}\Omega_i\Big)^{-1}\Big(\Sigma_\theta^{-1}\mu_\theta + \sum_{i=1}^{t-1}\Omega_i s_i\Big), \Big(\Sigma_\theta^{-1} + \sum_{i=1}^{t-1}\Omega_i\Big)^{-1}\Big)
\tag{36}
$$

The posterior distribution in Equation (36) applies for any $t > 0$. In particular, from the mean and covariance matrix it follows that at $t = 1$, when the firm has no prior export experience, its beliefs about export profitability reduce to:

$$\begin{aligned} \theta|d^0 &\sim \mathbb{N}\Big(\Big(\Sigma_\theta^{-1}\Big)^{-1}\Big(\Sigma_\theta^{-1}\mu_\theta\Big), \Big(\Sigma_\theta^{-1}\Big)^{-1}\Big) \\ &= \mathbb{N}\Big(\mu_\theta, \Sigma_\theta\Big) \end{aligned} \tag{37}$$

Thus, when a firm has no export experience, its beliefs coincide with the prior distribution of θ, which is assumed to be common knowledge. More generally, a firm's beliefs at period t about its export profitability, conditional on its export history d^{t-1}, are given by:

$$\Gamma_t(d^{t-1}) = \begin{cases} (\mu_\theta, \Sigma_\theta) & \text{if } t = 1 \\ (\mu_\theta^p, \Sigma_\theta^p) & \text{if } t > 1 \end{cases} \tag{38}$$

where:

$$\begin{aligned} \Omega_i &= \text{diag}((1/\sigma_\epsilon^2)d_i) \\ \mu_\theta^p &= \Big(\Sigma_\theta^{-1} + \sum_{i=1}^{t-1}\Omega_i\Big)^{-1}\Big(\Sigma_\theta^{-1}\mu_\theta + \sum_{i=1}^{t-1}\Omega_i s_i\Big) \\ \Sigma_\theta^p &= \Big(\Sigma_\theta^{-1} + \sum_{i=1}^{t-1}\Omega_i\Big)^{-1} \end{aligned} \tag{39}$$

Recall that we use Γ_t to denote the firm's learning process in Equation (23), since the information set in period t corresponds to its updated beliefs about true export profitability, given its export history up to that point. This links the learning process to the firm's discrete choice problem: in each period t, the firm selects export status d_t to maximize Equation (24). To evaluate expected profits in Equation (23), the firm relies on its posterior beliefs, which evolve according to Equation 39 as a function of export history d^{t-1}.

3.4 Dynamic Exporter Behavior

Although we initially described the firm's problem as static, the learning process makes clear that export decisions have dynamic implications. Entering a market not only affects current profits but also updates beliefs about profitability across all destinations, thereby shaping future entry and exit decisions.

Empirical evidence shows that some firms reenter export markets after exiting. Since a firm's precision about its true export profitability θ is non-decreasing with tenure, one way to rationalize this behavior is to allow fixed export costs to vary randomly. Following Cebreros (2016), we assume that the fixed cost of exporting to market j at time t is constant but subject to a random shock:

$$f_{jt} = f_j + \eta_{jt} \quad \text{where } \eta_{jt} \overset{\text{i.i.d.}}{\sim} \mathbb{N}(0, \sigma_\eta^2) \tag{40}$$

Using Equation (24), expected profits in t given export status d_t will now be given by:

$$\tilde{\Pi}(d_t|\Gamma_t) - \sum_{j \in D} d_{jt}\eta_{jt} \tag{41}$$

where η_{jt} is observed by the firm for all $j \in D$ before choosing export status d_t. The firm's recursive entry-exit problem can then be characterized by the following Bellman equation:

$$\mathbb{V}_\Psi(\mu_\theta^p, \Sigma_\theta^p, \eta) = \max_{d \in \Xi}\{\tilde{\Pi}(d|\mu_\theta^p, \Sigma_\theta^p) - \sum_{j \in D} d_j\eta_j + \beta\mathbb{E}_{\mu_\theta^{p'}, \Sigma_\theta^{p'}, \eta'}[\mathbb{V}_\Psi(\mu_\theta^{p'}, \Sigma_\theta^{p'}, \eta')]\} \tag{42}$$

subject to the evolution of the firm's beliefs about export profitability given by Equation (39). In this recursive formulation, Ξ denotes the set of all possible export status vectors d, β is the discount factor, and Ψ the vector of model parameters.

As discussed in Section 2.3.2, firms are forward-looking, so exporting to a destination is valuable not only for its expected profits $\tilde{\Pi}(d \mid \mu_\theta^p, \Sigma_\theta^p)$ but also for the information it provides about other markets. This informational value is captured in the continuation term of the Bellman equation. Because entry into j updates beliefs about profitability across all destinations, it directly influences the firm's decision to enter market j.

4 Quantifying the Model

Estimating the model in Section 3 is computationally demanding because the firm's state space includes its beliefs about profitability in all export markets —a vector of means and a covariance matrix —along with shocks to fixed export costs. If a Mexican firm can export to D destinations, the number of state variables equals $2D + D(D+1)/2$.

This is challenging for two reasons. First, the dynamic nature of the model requires value function iteration for estimation. Even with a relatively small set of ten potential export destinations, the state space would involve 75 variables, making value function iteration computationally infeasible. Second, the model is a combinatorial discrete choice problem:

firms must select the optimal combination of destinations from the set of all possible subsets, which grows exponentially with the number of export markets.

We address these challenges as follows. First, we restrict Mexican firms to at most five export destinations ($D = 5$), chosen as the top five in the customs data: the United States, Canada, Spain, Germany, and China. Second, we abstract from iceberg-type trade costs, which makes entry decisions independent across destinations. Under this assumption, the expression for expected total profits in Equation (22) simplifies to:

$$\tilde{\Pi}(d_t|\Gamma_t) = \Big\{\frac{1}{\sigma}\Big(\frac{\sigma-1}{\sigma}\Big)^{\sigma-1} - f\Big\} + \sum_{j\in D} d_{jt}\Big\{\frac{1}{\sigma}\Big(\frac{\sigma-1}{\sigma}\Big)^{\sigma-1}\mathbb{E}_{s_t}\big(h(s_{jt})^{\sigma}|\Gamma_t\big) - f_j\Big\} \tag{43}$$

This formulation implies that the entry problem is separable across destinations. Finally, we assume firms are myopic with respect to information spillovers: when deciding whether to enter a destination, they do not recognize how this choice affects beliefs about other markets, nor how experience in similar destinations informs profitability in the current one. Under this assumption, the firm's decision problem reduces to three state variables: the mean and variance of export profitability, and the current shock to fixed export costs (which is observable to the firm). The drawback is that the model's quantification will not capture the forward-looking behavior described in Section 2.3.2, where firms may enter destinations partly for their informational value.

4.1 Independent Export Destinations and Myopic Firms

We characterize the firm's optimization problem and derive belief expressions under the assumptions that entry decisions are independent across markets and that firms are myopic with respect to information spillovers. For clarity, in this section we drop subscript notation and instead use a recursive formulation to distinguish between current and future values of variables.

Following Equation (42), a firm's Bellman equation in a particular export market is:

$$\mathbb{V}_{\Psi}(\mu,\Sigma,\eta) = \max_{d\in\{0,1\}}\{\tilde{\Pi}(d|\mu,\Sigma) - d\eta + \beta\mathbb{E}_{\mu',\Sigma',\eta'}[\mathbb{V}_{\Psi}(\mu',\Sigma',\eta')]\} \tag{44}$$

where μ and μ' denote current and future beliefs about the expected value of export profitability, Σ and Σ' represent current and future beliefs about its variance, and η and η' are the current and future realizations of the fixed export cost shock.

Expected profits of exporting to a given destination are given by:

$$\tilde{\Pi}(d|\mu,\Sigma) = d\Big\{\frac{1}{\sigma}\Big(\frac{\sigma-1}{\sigma}\Big)^{\sigma-1}\mathbb{E}_s\big(h(s)^\sigma|\mu,\Sigma\big) - f\Big\} \tag{45}$$

As shown in Equation (25), a firm's profits in a given export destination depend on signal s, which combines true export profitability with noise. Firms observe this signal through realized export profits when they choose to export in a given period.

To form expectations about signal s —and thus about expected profitability when deciding whether to export to a destination —recall that the signal received by each firm is given by:

$$s = \theta + \epsilon \tag{46}$$

where we assume $\theta \sim \mathbb{N}(\mu_\theta, \sigma_\theta^2)$ and $\epsilon_{jt} \sim \mathbb{N}(0, \sigma_\epsilon^2)$, with both distributions known to the firm. The expected value of signal s, conditional on the firm's current beliefs about the mean μ and variance Σ of export profitability θ, is:

$$\begin{aligned} \Rightarrow \mathbb{E}(s|\mu,\Sigma) &= \mathbb{E}(\theta+\epsilon|\mu,\Sigma) \\ &= \mathbb{E}(\theta|\mu,\Sigma) + \mathbb{E}(\epsilon|\mu,\Sigma) = \mu + 0 \end{aligned} \tag{47}$$

while the variance of the signal, again conditional on a firm's current beliefs, is given by:

$$\begin{aligned} \Rightarrow \mathbb{V}(s|\mu,\Sigma) &= \mathbb{V}(\theta+\epsilon|\mu,\Sigma) \\ &= \mathbb{V}(\theta|\mu,\Sigma) + \mathbb{V}(\epsilon|\mu,\Sigma) \\ &= \Sigma + \sigma_\epsilon^2 \end{aligned} \tag{48}$$

Using the assumption that θ and ϵ are independent. The observed signal is believed by firms to be distributed according to:

$$s|\mu,\Sigma \sim \mathbb{N}(\mu, \Sigma + \sigma_\epsilon^2) \tag{49}$$

To compute the expected continuation value in Equation (44), we need to characterize the distribution of next period's beliefs (μ', Σ') given current beliefs. When export destinations are independent, Equation (39) implies that:

$$\mu = \Big(\frac{1}{\sigma_\theta^2} + \frac{n}{\sigma_\epsilon^2}\Big)^{-1}\Big(\frac{\mu_\theta}{\sigma_\theta^2} + \frac{1}{\sigma_\epsilon^2}\sum_{i=1}^{t-1} d_i s_i\Big) \tag{50}$$

$$\Sigma = \Big(\frac{1}{\sigma_\theta^2} + \frac{n}{\sigma_\epsilon^2}\Big)^{-1} \tag{51}$$

The updated covariance matrix Σ is deterministic, as it depends only on the known parameter Σ_θ and the number of periods n that the firm has previously exported to a destination. Given a realization of s' in the next period, the updated mean is:

$$\mu' = \Big(\frac{1}{\sigma_\theta^2} + \frac{n+d}{\sigma_\epsilon^2}\Big)^{-1}\Big(\frac{\mu_\theta}{\sigma_\theta^2} + \frac{1}{\sigma_\epsilon^2}\sum_{i=1}^{t-1} d_i s_i + \frac{d}{\sigma_\epsilon^2}s'\Big) \tag{52}$$

Note that if a firm does not export in the current period ($d = 0$), its beliefs remain unchanged, i.e., $\mu' = \mu$. Since s' is only observed in the following period, the expected value of tomorrow's mean belief, conditional on today's beliefs and export choice, is:

$$\mathbb{E}(\mu'|\mu, n, d) = \Big(\frac{1}{\sigma_\theta^2} + \frac{n+d}{\sigma_\epsilon^2}\Big)^{-1}\Big(\frac{\mu_\theta}{\sigma_\theta^2} + \frac{1}{\sigma_\epsilon^2}\sum_{i=1}^{t-1} d_i s_i + \frac{d}{\sigma_\epsilon^2}\mathbb{E}(s'|\mu, n)\Big) \tag{53}$$

which using Equation (49) gives us:

$$\mathbb{E}(\mu'|\mu, n, d) = \Big(\frac{1}{\sigma_\theta^2} + \frac{n+d}{\sigma_\epsilon^2}\Big)^{-1}\Big(\frac{\mu_\theta}{\sigma_\theta^2} + \frac{1}{\sigma_\epsilon^2}\sum_{i=1}^{t-1} d_i s_i + \frac{d}{\sigma_\epsilon^2}\mu\Big) \tag{54}$$

The variance of the updated beliefs for the mean μ' is given by:

$$\begin{aligned}\mathbb{V}(\mu'|\mu, n, d) &= \Big(\frac{d}{\sigma_\epsilon^2}\Big)^2\Big(\frac{1}{\sigma_\theta^2} + \frac{n+d}{\sigma_\epsilon^2}\Big)^{-2}\mathbb{V}(s'|\mu, n) \\ &= \Big(\frac{d}{\sigma_\epsilon^2}\Big)^2\Big(\frac{1}{\sigma_\theta^2} + \frac{n+d}{\sigma_\epsilon^2}\Big)^{-2}\Big(\Big(\frac{1}{\sigma_\theta^2} + \frac{n}{\sigma_\epsilon^2}\Big)^{-1} + \sigma_\epsilon^2\Big) \end{aligned} \tag{55}$$

Note that if a firm chooses not to export in the current period ($d = 0$), then $\mathbb{V}(\mu' \mid \mu, n) = 0$. This result is intuitive: since no new information is obtained, beliefs remain unchanged at μ, leaving no uncertainty about the future value of μ'.

4.2 Parametrization of the Model

We use the following functional form for demand shifter $h(\cdot)$ in Equation (23):

$$h(s) = \kappa \exp(-\lambda \exp(-gs)) \quad \text{where} \quad \kappa, \lambda, g > 0 \tag{56}$$

where κ sets the upper bound of the $h(\cdot)$ function, while λ and g govern its growth rate. This parametrization ensures that $h(\cdot)$ is continuous, differentiable, bounded, positive, and monotone.

Table 5 reports the parameter values used in the model. The elasticity of substitution σ and the parameters λ and g of the $h(\cdot)$ function are taken from the literature, with estimates from Cebreros (2016), who also rely on Mexican data. The discount factor is set to $\beta = 0.95$, and κ is set to 2.5 so that the model replicates observed firm-level exports while ensuring that exports remain bounded. Finally, the variance of shocks to export profitability is set to $\sigma_\epsilon^2 = 0.23$, a value chosen to match the observed variance of export revenues.

Parameter	**Value**	**Source**
β	0.95	-
σ	3.85	Antras et al. (2017)
λ	2.64	Cebreros (2016)
g	2.69	Cebreros (2016)
κ	2.50	Calibrated
σ_ϵ^2	0.23	Calibrated

Table 5: Parameters from the literature.

We assume that the correlation in export profitability between two countries is given by the import-based similarity index $S(i, j) \in [0, 1]$ introduced in Section 3.3. Thus, the greater the similarity between two destinations, the higher the correlation in firms' profitability across them. Using these correlation coefficients together with our variance estimates, we recover the covariances of export profitability across country pairs and construct the full covariance matrix Σ_θ. This matrix is then used in simulating the model and conducting counterfactual exercises.

Finally, to reduce the computational burden of estimation, we assume that the fixed cost shock η in Equation (44) takes only two values, $-\bar{\eta}$ or $\bar{\eta}$, each with equal probability. As discussed earlier, this shock is necessary to rationalize why some firms may reenter an export market after having exited.

4.3 Estimation of Export Profitability and Fixed Costs

This section outlines the structural estimation of firms' export profitability and fixed costs, conducted separately for each destination as described in Section 4.1. Estimation relies on the Nested Fixed-Point (NFXP) algorithm introduced by Rust (1987). The procedure consists of two loops: an outer loop that carries out the estimation routine and an inner loop that solves the firm's problem for each candidate parameter vector.

In the outer loop, we employ the Simulated Method of Moments (SMM) to identify the parameter vector that best aligns model simulations with moments observed in the Mexican customs data. We use SMM because the model does not yield closed-form solutions for firm behavior, making simulation necessary. Let x denote the data and ξ a candidate parameter vector. Our implementation of SMM solves for the ξ that minimizes the following objective function:

$$\min_{\Psi} ||e(\tilde{x}, x|\Psi)||$$

The moment error function $e(\tilde{x}, x|\Psi)$ is expressed as the percent difference between the vectors of simulated and data moments:

$$e(\tilde{x}, x|\Psi) = \frac{\hat{m}(\tilde{x}|\Psi) - m(x)}{m(x)}$$

where $m(\cdot)$ denotes a set of R distinct moments, and $\tilde{x}$ is the simulated data generated from the model under parametrization Ψ. Using the L^2 norm, our SMM implementation seeks the parameter vector that minimizes the sum of squared errors:

$$\hat{\Psi} = \arg\max_{\Psi} e(\tilde{x}, x|\Psi)^T I_R e(\tilde{x}, x|\Psi)$$

where I_R is an $R \times R$ identity matrix. We define the objective function in terms of percentage deviations, ensuring that all moments in $m(\cdot)$ are expressed in comparable units and that no moment is unintentionally over-weighted.

The vector of parameters we estimate using SMM for each of the export destinations included in our quantification is the following:

$$\Psi = [\mu_\theta, \sigma_\theta^2, f, \bar{\eta}]$$

Specifically, the parameters to be estimated are the mean and variance of export profitability, the fixed cost of exporting, and the magnitude of the fixed cost shock. The set of moments used to identify the parameter vector $\hat{\Psi}$ are:

1. Average log exports at the firm level: This moment identifies the mean of export profitability μ_θ, since higher values of μ_θ generate larger signals s and thus higher export revenues.

2. Standard deviation of log exports at the firm level: This moment identifies the variance of export profitability σ_θ^2, since higher variance in θ increases the dispersion of signals s and, consequently, the variability of export revenues.

3. Share of firms exporting: This moment identifies the fixed cost of exporting f, since higher values of f reduce the share of firms able to cover fixed costs with operating profits.

4. Average length of export spells: This moment identifies the shock to fixed costs of exporting $\bar{\eta}$, since larger values of $\bar{\eta}$ allow firms to withstand adverse shocks to signal s, prolonging export spells.

Second, for the inner loop of the estimation algorithm, we implement Value Function Iteration. Given a parameter vector Ψ and an arbitrary initial guess $\mathbb{V}_\Psi^0(\cdot)$, Equation (44) can be solved recursively as:

$$\mathbb{V}_\Psi^{t+1} = \max_{d\in\{0,1\}}\{\tilde{\Pi}(d) - d\eta + \beta\mathbb{E}[\mathbb{V}_{\Psi_j}^t]\} \tag{57}$$

where superscript t indexes the iteration. Since $h(\cdot)$ is continuous and bounded, expected profits $\tilde{\Pi}(d)$ inherit these properties. Consequently, after a sufficiently large number of iterations, the algorithm converges to a fixed point where $\mathbb{V}^{t+1}\Psi = \mathbb{V}^t\Psi$.

At each iteration, we compute $\mathbb{V}_\Psi^t$ over the entire state space, where a firm's state variables are its current belief about mean profitability μ, its export tenure n, and the current fixed cost shock $\bar{\eta}$. As shown in Equations (49), (51), and (52), these variables are sufficient to calculate $\tilde{\Pi}(d)$ and the expected continuation value in Equation (57). The firm's only control variable is its entry decision d into the export market.

We discretize the state space using grids of 50 values for μ, 20 values for n (implicitly assuming firms can export for up to 20 periods), and two values for the fixed cost shock, $-\bar{\eta}$ and $\bar{\eta}$. These grids are constructed separately for each parameter vector Ψ in the outer loop.

We compute $\tilde{\Pi}(d)$ via Monte Carlo integration and build transition probability matrices using Equations (54) and (55) to evaluate the expected continuation value in Equation (57).

After convergence to a fixed point of the value function, we simulate firm behavior. For a given parameter vector Ψ, we draw 10,000 realizations of firm profitability from $\mathbb{N}(\mu_\theta, \sigma_\theta^2)$ and simulate firms' entry and exit decisions over 20 periods. In each period, firms decide whether to export; if they do, they observe a signal s —a random draw —and update beliefs as described in Section 4.1. From these simulations we compute the moments $\hat{m}(\tilde{x} \mid \Psi)$, which are then compared with data moments $m(x)$. If the distance between the two sets of moments falls within a specified tolerance, we identify $\hat{\Psi}$.

Estimation results are reported in Table 6. The model suggests that, on average, Canada, China, and Spain offer similar export profitability for Mexican firms. By contrast, the United States is estimated to be significantly more profitable, consistent with its large market size and consumers' relatively strong preferences for Mexican varieties. Variance estimates are highest for Canada and the United States. Since these are the main destinations for Mexican exporters, the larger heterogeneity among firms serving these markets translates into higher estimated variance of export profitability.

Parameter	**Canada**	**China**	**Germany**	**Spain**	**USA**
μ_θ	0.96	0.99	0.85	0.96	1.25
σ_θ^2	0.33	0.29	0.27	0.21	0.35
f	3.34	3.04	2.61	2.95	0.39
$\bar{\eta}$	0.95	0.42	0.61	0.72	0.90

Table 6: SMM Estimated Parameters.

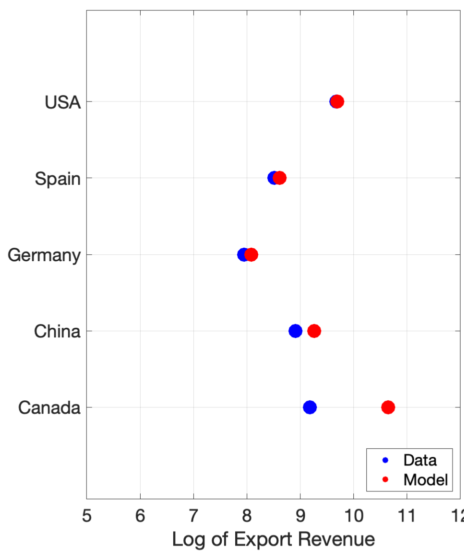
For both the fixed cost of exporting and its shock, units are expressed in log USD. The estimates indicate that exporting to the United States is substantially less costly for Mexican firms than exporting to other destinations. Combined with the higher mean profitability, this explains why most Mexican firms export to the US at some point, while entry rates into other destinations remain much lower. Finally, the estimated shocks to fixed export costs are smaller for China and Germany, consistent with the shorter average export spells observed in these markets.

As described in Section 4.2, we proxy the correlation coefficients of export profitability using the similarity index across destinations. Combined with the variance estimates, this yields the covariance matrix of export profitability, $\hat{\Sigma}_\theta$, reported in Table 7. The results indicate that the largest covariances —and thus the strongest information spillovers —occur between the USA and Canada, Canada and Germany, and the USA and China. By contrast,

	Canada	China	Germany	Spain	USA
Canada	0.33	0.16	0.18	0.06	0.27
China	0.16	0.29	0.11	0.08	0.20
Germany	0.18	0.11	0.27	0.04	0.15
Spain	0.06	0.08	0.04	0.21	0.07
USA	0.27	0.20	0.15	0.07	0.35

Table 7: Estimated Covariance Matrix of Export Profitability.

(a) Average firm-level log exports



(b) Standard deviation of log exports

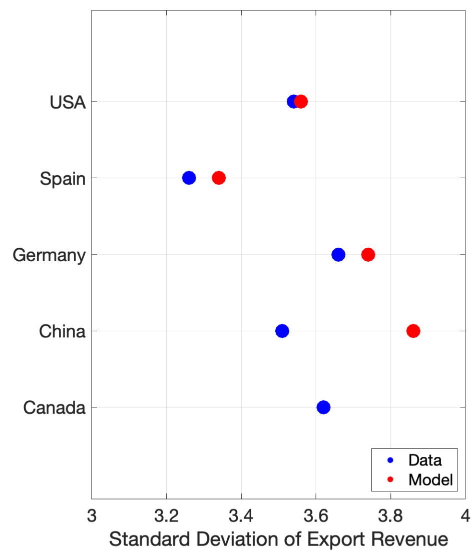


Figure 2: Fit of the model regarding the mean and variance of log exports.

the covariance between Germany and Spain is surprisingly low, suggesting that despite their geographic proximity, these countries may exhibit markedly different preferences for Mexican varieties.

Figures 2 and 3 compare selected moments from the data with those generated by our simulation to assess model fit. Overall, the model performs well, with two notable exceptions: the standard deviation of log exports for China and the average length of export spells in the USA. The first discrepancy likely reflects an overestimation of China's variance in export profitability, while the second suggests that the model may overstate the magnitude of fixed cost shocks in the US market.

(a) Share of firms exporting (b) Average length of export spells

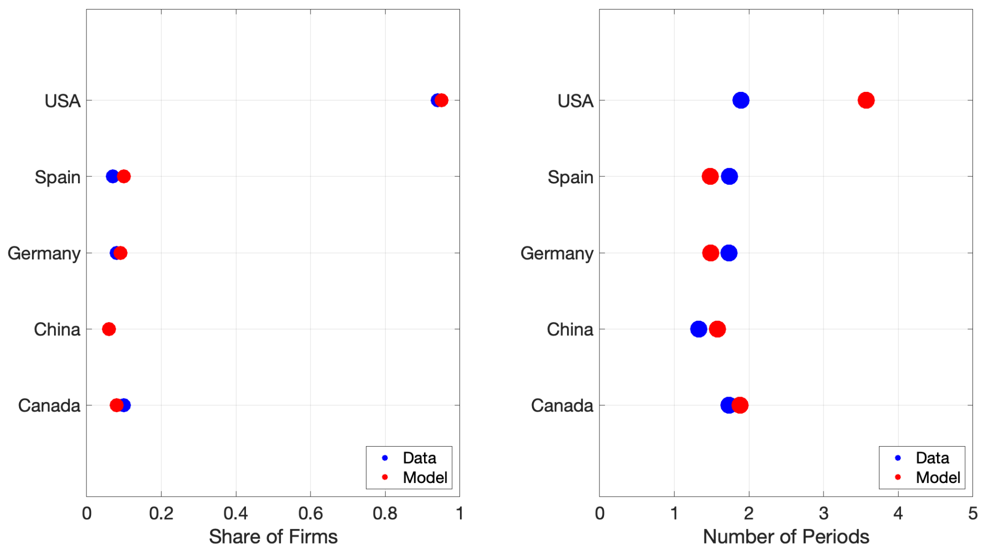


Figure 3: Fit of the model regarding entry shares and export spells.

4.4 Evolution of Entry Shares and Beliefs

To illustrate the mechanics of our model, Figures 4 and 5 present simulation results. Panel 4a plots the share of firms exporting to each destination against tenure in that market, showing that firms rarely continue exporting to a destination for many periods after entry. Panel 4b displays the corresponding model simulation, which replicates the declining relationship between entry shares and tenure. In the model, initial entry is often driven by a favorable shock to fixed export costs, but as firms learn about their profitability, many choose to exit. The exception is the United States: the model fails to match the observed decline, instead predicting that few firms stop exporting there over time.

Figure 5 illustrates how average export profitability and beliefs evolve over time. Panel 5a reports results for Canada, China, and Germany, while Panel 5b shows Spain and the USA. Solid lines represent firms' beliefs about expected profitability, and dashed lines represent actual profitability. Although true export profitability is constant over time, the composition of exporting firms changes. As less profitable firms exit, the remaining exporters are those with higher profitability, causing average observed profitability to rise with tenure, as also reflected in Panel 4b.

The dashed lines in Figure 5 need not coincide with the estimated means in Table 6, since the former represent mean profitability conditional on exporting, while the latter report unconditional means. The figure also illustrates the link between export spells and

(a) Data for the share of firms exporting (b) Simulation for the share of firms exporting

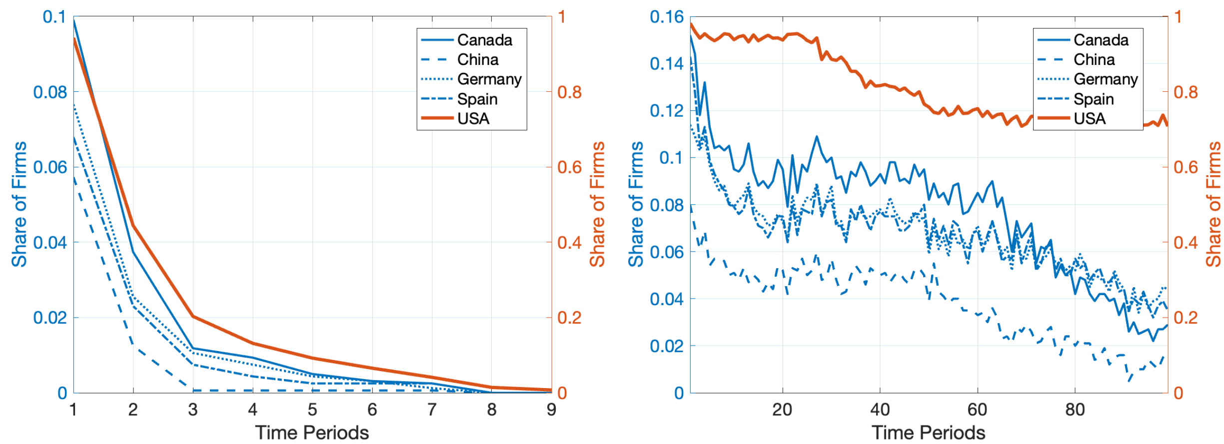


Figure 4: Data and simulation of the model for the share of firms exporting.

signal precision: as firms gain experience, uncertainty declines and beliefs converge to true profitability, causing the solid and dashed lines to draw closer over time. Variation in the set of exporting firms affects both average profitability and beliefs, but beliefs also evolve due to noise in profitability signals. For instance, a profitable firm may receive a negative demand shock, leading it to believe exporting is unprofitable and to exit the market. Re-entry would only occur if a positive fixed-cost shock encouraged it to try again, allowing the firm to acquire additional information and potentially discover its profitability.

5 Counterfactual Scenarios

In this section, we conduct two sets of counterfactual simulations. The first examines the role of information spillovers in export diversification. Specifically, we remove information spillovers and, using the parameter estimates from Section 4, compare key model moments to those from the baseline simulation with spillovers. The second set evaluates the impact of reducing fixed export costs, focusing on how these changes affect the average number of destinations served by firms. Such cost reductions could arise from policies that facilitate firms' participation in international trade or from free trade agreements that lower entry costs into foreign markets.

The second set of counterfactuals suggests a potential second-order effect of free trade agreements: greater diversification of export destinations. We emphasize the word potential because information spillovers do not necessarily increase the share of firms exporting to any given destination. As discussed in Section 2, the role of learning is to improve firms' information about demand conditions abroad, enabling them to make more informed entry

(a) Simulation for export profitability and beliefs (b) Simulation for export profitability and beliefs

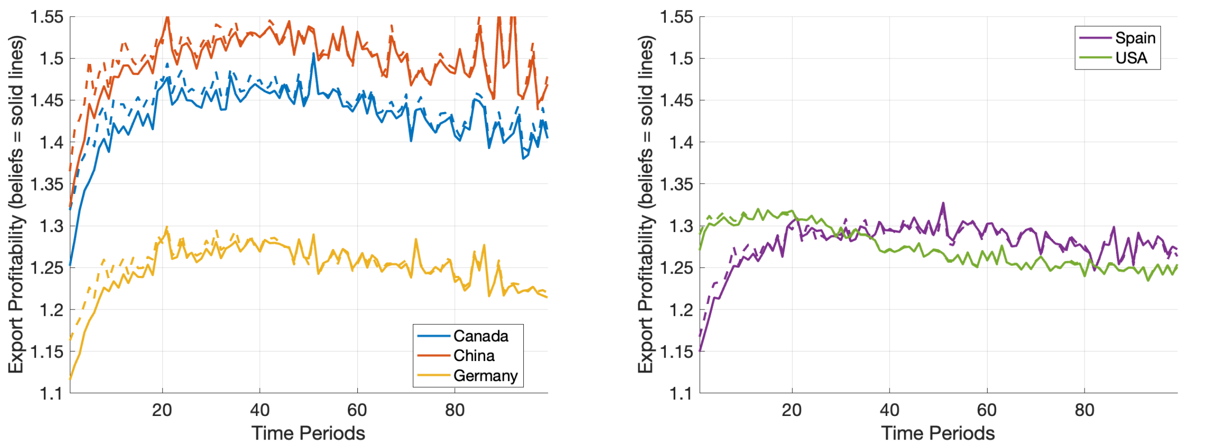


Figure 5: Simulation of the model for export profitability and firms' beliefs.

and exit decisions.

5.1 Removing Information Spillovers

We remove information spillovers by assuming that, although firms' export profitabilities remain correlated across destinations, firms are unaware of this correlation. Concretely, we retain the covariance structure reported in Table 7, but assume it is not common knowledge when firms form beliefs about export profitability. Under this assumption, a firm's prior beliefs about profitability, before gaining any export experience, are distributed as:

$$\theta | d^0 \sim \mathbb{N}(\hat{\mu}_\theta, \tilde{\Sigma}_\theta) \quad \text{where } \tilde{\Sigma}_\theta \equiv I_D \odot \hat{\Sigma}_\theta \tag{58}$$

In practice, although export profitability remains correlated across destinations under $\hat{\Sigma}\theta$, firms behave as if profitability were distributed according to the diagonal covariance matrix $\tilde{\Sigma}\theta$. As a result, they do not learn about other markets when exporting to a given one. To implement this counterfactual, we simulate a large number of firms following the procedure in Section 4.3, but assume all firms hold prior beliefs given by Equation (58). Figure 6 reports the results, comparing key model moments from the baseline simulation with those obtained under this alternative specification of the prior covariance matrix.

Without information spillovers, the share of firms exporting to Canada, China, and Germany would fall by 26.6%, 45.4%, and 22.3%, respectively. By contrast, the shares exporting to Spain and the United States would rise by 8.1% and 25.2%. These results underscore the central role of the US in the diversification of Mexican exports: it is through

Figure 6: Effects of removing information spillovers.

exporting to the US that Mexican firms acquire information about their profitability in other foreign markets, enabling some to expand into those destinations.

The intuition behind these results is as follows. As shown in Table 6, the fundamentals of export profitability represent firms' prior beliefs in the absence of experience. Under these priors, destinations such as Canada, China, and Germany are not profitable for the average firm. When spillovers are removed, firms only learn about profitability in the specific markets they enter. Thus, sustained exporting occurs only when a profitable firm —one with a sufficiently high draw of export profitability —also experiences a favorable shock to fixed costs. This leads to a much slower learning process compared to the spillover case, where firms can enter the US —a profitable destination on average —and use that experience to learn about non-US markets.

Average firm-level export revenue, conditional on exporting, declines in all destinations except Spain. This pattern reflects changes in the composition of firms that continue exporting. Without information spillovers, firms cannot learn about their profitability from exporting to other markets. As a result, entry into destinations such as China or Germany is largely driven by positive shocks to fixed export costs —shocks that are independent of a firm's underlying profitability in those markets.

Lastly, the average length of export spells falls in Canada and Germany, consistent with entry into these destinations being driven mainly by fixed-cost shocks, which shorten firms' time in the market. Specifically, spell length decreases by 5.3% in Canada and 10.8% in Germany. By contrast, average export spells increase in China (1.4%), Spain (9.9%), and especially the US (93.2%). The large increase in US spell length arises because fewer firms export to alternative destinations, slowing the learning process: it takes longer for firms' beliefs to converge to their true profitability. In the US, many firms enter due to its high mean profitability and low fixed costs, but some require more time to discover that they are not profitable in this market.

Overall, this counterfactual highlights the critical role of information spillovers in shaping export diversification. The results suggest that entry into smaller, less profitable markets is largely driven by the experience Mexican firms acquire while exporting to the US. On average, without information spillovers, the number of non-US destinations served by Mexican firms would be 21.6% lower, equivalent to a decline of 1.08 destinations per firm.

5.2 Reductions in the Fixed Costs of Exporting

This counterfactual quantifies the effects of lowering fixed costs of exporting (FCE) to a specific country. We set $f = 0$ for one destination at a time and simulate firms' entry and exit choices for the selected destinations. Intuitively, the results depend on which country experiences the cost reduction and its similarity to other destinations. Entry remains partly driven by shocks to FCE, $\bar{\eta}$, and we retain the covariance matrix of export profitability shown in Table 7.

Figure 7 presents the counterfactual results for average firm-level exports and the share of firms exporting. Each group of bars reports the effects across all destinations when fixed costs are eliminated for the country indicated on the y-axis. Panel 7a shows the impact on the share of firms exporting to each destination. As expected, this share rises sharply for the destination where fixed costs were reduced.

When FCE is reduced for Canada, the share of firms exporting to China, Germany, and Spain increases by 12.7%, 8.7%, and 2.9%, respectively. However, higher entry into one destination does not always translate into higher entry elsewhere; in fact, entry may fall, as in the case of China. Here, firms that discover they are unprofitable in China may infer that they are unlikely to be profitable in other markets, reducing entry. Lowering FCE for Germany raises entry into Spain by 43.6%, while reducing FCE for Spain raises entry into Germany by 62.9%. Although Germany and Spain are not especially similar in terms of estimated export profitability, their correlation is sufficient to generate higher cross-entry. By contrast, reducing FCE for the US does not increase entry there, since most Mexican firms already export to this market. Intuitively, because entry into the US is unaffected, firms do not acquire additional information that would encourage entry into other destinations.

Panel 7b reports the effects on firm-level exports. On average, exports fall when FCE

(a) Share of firms exporting

(b) Average firm-level exports

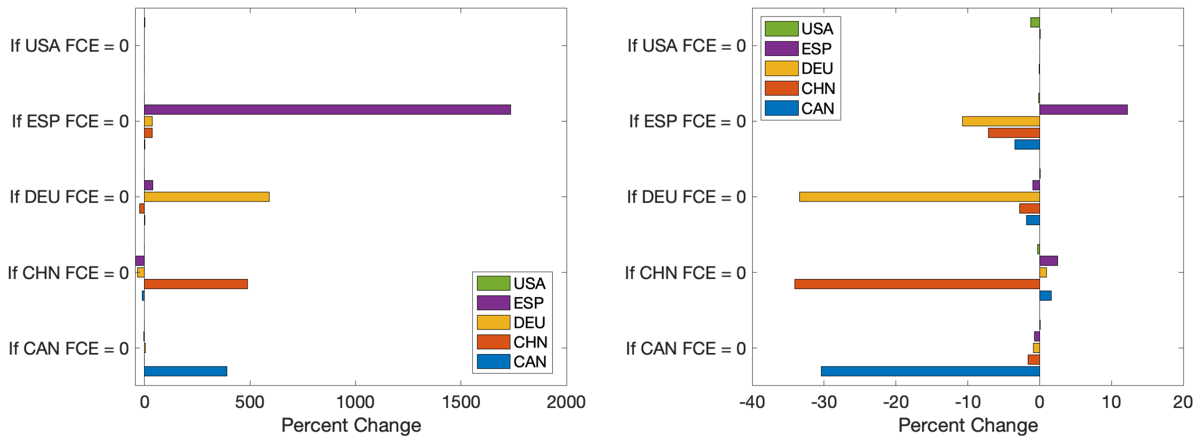


Figure 7: Effects of a reduction in fixed costs of exporting.

are reduced, reflecting a composition effect: lower fixed costs allow less profitable firms to enter, which reduces average exports. This effect is strongest in the country experiencing the cost reduction, since it sees the largest increase in entry. Panel 8a shows the impact on the average length of export spells. Overall, spell length rises alongside higher entry shares, both in the treated country and in other destinations. Greater entry improves the precision of profitability beliefs through information spillovers, enabling firms to make better entry decisions and sustain longer export spells.

Lastly, Panel 8b presents the effects on export diversification. To isolate spillover-driven diversification, we compute the average number of destinations per firm excluding the country where FCE was reduced; otherwise, results would be dominated by direct entry into that market. The findings show that reducing FCE to Canada and Spain increases diversification. Specifically, lowering FCE to Canada raises the average number of destinations by 0.14, while lowering FCE to Spain increases it by 0.61. When FCE are lowered for China, diversification decreases due to a decline in the share of firms exporting to destinations other than China. A reduction in Germany's FCE also slightly reduces diversification, as the fall in entry into Canada and China outweighs the rise in entry into Spain. Once again, lowering FCE to the US has no significant effect on diversification, since entry shares remain largely unchanged.

This counterfactual suggests that lowering FCE can increase both the share of exporting firms and the diversification of destinations. The magnitude of these effects depends on the similarity between countries and the information firms gain while exporting. For instance, adverse demand shocks in one destination may discourage entry into similar markets. Overall, the results suggest that policies facilitating international trade —such as reducing

(a) Average length of export spells

(b) Number of exported destinations

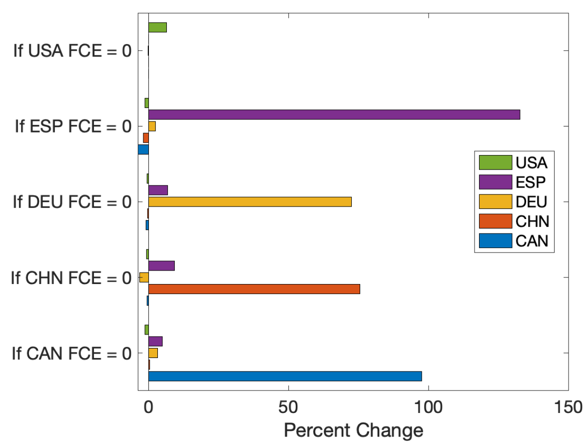


Figure 8: Effects of a reduction in fixed costs of exporting.

export costs or implementing multilateral free trade agreements —can foster greater export diversification, or at minimum, enable firms to make more informed choices about where to export.

6 Conclusions

This paper examines how information spillovers across foreign markets shape firms' entry and exit decisions. When demand conditions are correlated across destinations, firms update their beliefs about profitability based on export experience. Using data on the universe of Mexican exporters, we show empirically that the probability of making an incorrect entry decision —entering an unprofitable market —declines as firms accumulate relevant export experience. Conversely, because destinations provide information about other markets, the more informative a destination is, the more likely a firm is to enter it.

We develop and estimate a dynamic model of export supply with information spillovers. In the model, Mexican firms choose each period where to export, accumulate experience, and update beliefs about profitability in other markets, thereby shaping future entry decisions. We quantify firm-level profitability and fixed costs for five major destinations and then conduct counterfactual simulations. Without information spillovers, the share of firms exporting outside the US would fall by 21.6%, highlighting the critical role of the US as a source of information about other markets. A second set of counterfactuals shows that lowering fixed export costs to a particular destination increases export diversification and raises exports and export spell length in other markets.

The results of this paper underscore the importance of learning in foreign markets for firm behavior and export diversification. Trade policies that encourage participation in international markets can improve outcomes such as export volumes and export duration. By lowering entry barriers —through free trade agreements or government programs that incentivize firms to export —such policies provide firms with better information about their profitability, enabling them to make more informed entry decisions.

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